

Surname / Van

Initials / Voorletters

Student Number / Studentenomer

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staafkodeplakker
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FSK116

Semester Test 1 / Semestertoets 1

Department of Physics, University of Pretoria

Department Fisika, Universiteit van Pretoria

Date / Datum: 16 March 2018 / 16 Maart 2018

Examiner / Eksaminator : Mr M.J. Legodi & Prof W.E. Meyer

Duration / Tydsduur : 2 hours / uur

Points / Punte : 60

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TEST REGISTER: Carefully fold and tear along the dashed line and hand over strip to the invigilator
TOETSREGISTER: Vou en skeur noukeurig langs die stippellyn en handing die strokie by die toesighouer in.

Surname Van					Initials Voorletters		
Student Number Studentenommer							
Venue Lokaal							
Signature Handtekening							

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Plak staafkodeplakker hier!

Semester Test 1 / *Semestertoets 1*

Department of Physics, University of Pretoria

Department Fisika, Universiteit van Pretoria

Date / *Datum*: 16 March 2018 / *16 Maart 2018*

The following rules apply / *Die volgende reëls geld*:

All test and exam regulations of the University of Pretoria are applicable / *Alle toets- en eksamenregulasies van die Universiteit van Pretoria is van toepassing*.

Answer all questions / *Beantwoord alle vrae*.

Pocket calculators may be used, but not programmable calculators / *Sakrekenaars mag gebruik word, maar nie programmeerbare sakrekenaars nie*.

No cell phones are allowed in the examination venue / *Geen selfone word in die eksamenlokaal toegelaat nie*.

This paper may not be taken out of the examination venue / *Hierdie vraestel mag nie uit die eksamenlokaal geneem word nie*.

A student is not allowed to write on any paper except that provided for the purpose of the examination / *Geen student mag op enige ander papier skryf as dié wat vir die doeleindes van die eksamen verskaf is*.

Please provide labelled sketches, including co-ordinate systems, to accompany your answers. Failure to do so may result in marks being deducted / *Verskaf asseblief benoemde sketse, insluitend koördinaatstelsels, om u antwoorde te vergesel. Versuim om dit te doen, kan daartoe lei dat punte afgetrek word*.

$$g = 9.80 \text{ m/s}^2$$
$$1 \text{ rev} = 360^\circ = 2\pi \text{ rad}$$

Question 1 / Vraag 1

(i) Two vectors are given by:

$\vec{A} = (4.00 \text{ m}) \hat{i} - (3.00 \text{ m}) \hat{j} + (1.00 \text{ m}) \hat{k}$ and $\vec{B} = (-1.00 \text{ m}) \hat{i} + (1.00 \text{ m}) \hat{j} + (4.00 \text{ m}) \hat{k}$. In unit vector notation, find vector $\vec{C}$ such that $\vec{A} + \vec{B} + \vec{C} = \vec{0}$.

[5]

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Twee vektore word gegee deur:

$\vec{A} = (4.00 \text{ m}) \hat{i} - (3.00 \text{ m}) \hat{j} + (1.00 \text{ m}) \hat{k}$ en $\vec{B} = (-1.00 \text{ m}) \hat{i} + (1.00 \text{ m}) \hat{j} + (4.00 \text{ m}) \hat{k}$. In eenheidsvektornotasië, vind vektor $\vec{C}$ so dat $\vec{A} + \vec{B} + \vec{C} = \vec{0}$.

$$\begin{aligned}\vec{A} &= 4.00 \text{ m } \hat{i} - 3.00 \text{ m } \hat{j} + 1.00 \text{ m } \hat{k} \\ \vec{B} &= -1.00 \text{ m } \hat{i} + 1.00 \text{ m } \hat{j} + 4.00 \text{ m } \hat{k} \\ \vec{C} &= C_x \hat{i} + C_y \hat{j} + C_z \hat{k}\end{aligned}$$

$$\vec{A} + \vec{B} + \vec{C} = \vec{0}$$

$$\Rightarrow (4.00 - 1.00) \text{ m} + C_x = 0$$

$$\text{and } (-3.00 + 1.00) \text{ m} + C_y = 0$$

$$\text{and } (1.00 + 4.00) \text{ m} + C_z = 0$$

$$\therefore C_x = (-4.00 + 1.00) \text{ m} = -3.00 \text{ m} \checkmark$$

$$C_y = (-1.00 - 4.00) \text{ m} = -5.00 \text{ m} \checkmark$$

$$C_z = (3.00 - 1.00) \text{ m} = 2.00 \text{ m} \checkmark$$

$$\therefore \vec{C} = -3.00 \text{ m } \hat{i} - 5.00 \text{ m } \hat{j} + 2.00 \text{ m } \hat{k} \checkmark \checkmark$$

(ii) The speed of a sound wave is given by the equation $v = \sqrt{B/\rho}$, where B is the bulk modulus of the medium (measured in N/m^2) and ρ is the density of the medium. Write down the units of B and ρ in terms of S.I. base units. [2]

Die spoed van 'n klankgolf word gegee deur die vergelyking $v = \sqrt{B/\rho}$, waar B die massamodulus van die medium (gemeet in N/m^2) en ρ die digtheid van die medium is. Skryf die eenhede van B en ρ in terme van S.I.-basiseenhede neer.

B : $\text{kg m}^{-1} \text{s}^{-2}$ ✓ or kg/m/s^2 or $\text{kg}/(\text{m/s}^2)$ or in words
 But NOT N/m^2 nor N/m^2 N m^{-2}
 ρ : kg/m^3 or kg m^{-3} or in words ✓

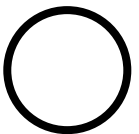
2

(iii) Show that v , as given by the equation above, has units of speed. [3]

Toon dat v , soos deur die vergelyking hier bo gegee, eenhede van spoed het.

Toon dat v bo eenhede van spoed het

$$[v] = \sqrt{B/\rho} = \sqrt{\frac{\text{kg m}^{-1} \text{s}^{-2}}{\text{kg m}^{-3}}} = \sqrt{\text{m}^2 \text{s}^{-2}} = \text{m/s} \quad \boxed{3}$$



Question 2 / Vraag 2

Utilize derivatives! A projectile is fired across a field with an initial velocity of 40 m/s. The horizontal position from the launch point is given by x and the vertical position (height) by y . The equation which describes the motion is given by: $y = -(x^2/20) + (x/\sqrt{3})$, where x and y are both in meters. (i) For what value of x will the projectile reach its maximum height? (ii) What is this maximum height? [5]

5

Gebruik afgeleides! 'n Projektiel word oor 'n horisontale veld afgevuur teen 'n beginspoed van 40 m/s. Die horisontale posisie vanaf die lanseerpunt (beginpunt) word gegee deur x , en die vertikale posisie (hoogte) deur y . Die vergelyking wat die baan van die projektiel beskryf, word gegee deur: $y = -(x^2/20) + (x/\sqrt{3})$, waarin beide x en y in meter gemeet word. (i) Vir watter waarde van x bereik die projektiel sy maksimum hoogte? (ii) Wat is hierdie maksimum hoogte?

$V_y = 40 \text{ m/s}$; (x, y) : position

$$y = \left(-\frac{x^2}{20}\right) + \left(\frac{x}{\sqrt{3}}\right), \text{ } x \text{ and } y \text{ in meters}$$

Taking the first derivative and equating it to zero.

$$\frac{dy}{dx} = \frac{2x}{20} + \frac{1}{\sqrt{3}} = 0 \text{ for maximum}$$

$$\therefore 2x = 20\left(\frac{1}{\sqrt{3}}\right)$$

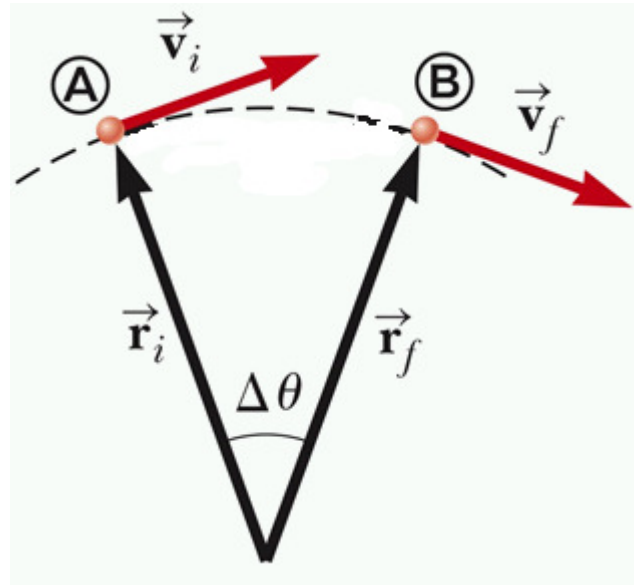
$$\Rightarrow x = \frac{10}{\sqrt{3}}$$

$$y_{\max} = \frac{\left(\frac{10}{\sqrt{3}}\right)^2}{20} + \frac{\left(\frac{10}{\sqrt{3}}\right)}{\sqrt{3}}$$
$$= 1.7 \text{ m}$$

5

Question 3 / Vraag 3

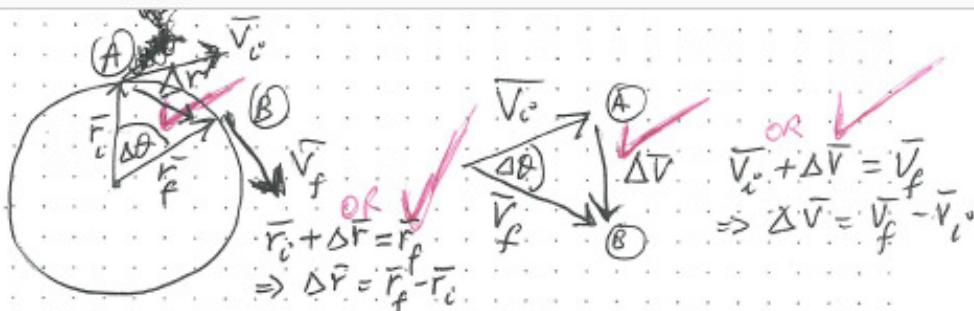
The figure on the right shows a particle undergoing uniform circular motion travelling from point A (position $\mathbf{r}_i$ and velocity $\mathbf{v}_i$) to point B (position $\mathbf{r}_f$ and velocity $\mathbf{v}_f$). Derive an expression for the magnitude of the centripetal acceleration in terms of r the radius of the circle and v the speed of the particle. **Hint:** Graphically determine $\Delta \mathbf{r}$ and $\Delta \mathbf{v}$ and show that they are related. Use this result to calculate the average acceleration of the particle. Explain your argument clearly. [5]



/5

Die figuur hier regs toon 'n deeltjie wat uniforme sirkelbeweging ondergaan en van punt A (posisie $\mathbf{r}_i$ en snelheid $\mathbf{v}_i$) na punt B (posisie $\mathbf{r}_f$ en snelheid $\mathbf{v}_f$). Lei 'n uitdrukking af vir die grootte van die sentripitale versnelling in terme van r , die radius van die sirkel en v die snelheid van die deeltjie. Woorde: Bepaal grafies $\Delta \mathbf{r}$ en $\Delta \mathbf{v}$ en vind

'n
ber



The motion is uniform circular motion since the particle has a constant speed around its path.
The particle has velocity $\mathbf{v}_i$ and position vector $\mathbf{r}_i$ at A and velocity $\mathbf{v}_f$ and position vector $\mathbf{r}_f$ at B.

$$v_f = v_i = v \text{ and}$$

$$r_f = r_i = r \text{ since we have uniform circular motion.}$$

The velocity triangle and the position vector triangle are similar because they share angle $\Delta \theta$ and the ratios between their sides are equal.

$$\therefore \frac{|\Delta \mathbf{v}|}{v} = \frac{|\Delta \mathbf{r}|}{r} \Rightarrow \Delta \mathbf{v} = \frac{v}{r} |\Delta \mathbf{r}|$$

The average acceleration between A and B is given by:

$$|\bar{\mathbf{a}}_{\text{avg}}| = \frac{|\Delta \mathbf{v}|}{|\Delta t|} = \frac{v}{r} \frac{|\Delta \mathbf{r}|}{\Delta t}$$

As $\Delta t \rightarrow 0$ when B approaches A,

$$|\bar{\mathbf{a}}_{\text{avg}}| = \frac{v}{r} \frac{dr}{dt} = \frac{v^2}{r}$$

$$\Rightarrow a_c = \frac{v^2}{r}$$

Question 4 / Vraag 4

An athlete swings a ball, connected to the end of a chain, in a horizontal circle. The athlete is able to rotate the ball at the rate of 8.00 rev/s when the length of the chain is 0.600 m. On increasing the length of the chain to 0.900 m, he is now able to rotate the ball to only 6.00 rev/s. [10]

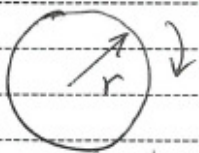
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- (a) Which rate of rotation gives the greater speed for the ball? (4)
 (b) What is the centripetal acceleration of the ball in the two circumstances, (i) at 8.00 rev/s and (ii) at 6.00 rev/s? (3)
 (c) How many degrees per minute are covered when the ball is rotating at 8.00 rev/s? (3)

'n Atleet swaai 'n bal wat aan die end van 'n ketting vasgemaak is, in 'n horisontale sirkel. Die atleet kan die bal teen 8.00 rev/s roteer wanneer die lengte van die ketting 0.600 m is. As hy die lengte van die ketting tot 0.900 m verhoog, kan hy die bal teen net 6.00 rev/s swaai.

- (a) Watter rotasiesnelheid gee die hoër spoed vir die bal?
 (b) Wat is die sentripetale versnelling van die bal in die twee gevalle, (i) teen 8.00 rev/s en (ii) teen 6.00 rev/s?
 (c) Hoeveel grade per minuut word gedek wanneer die bal teen 8.00 rev/s draai?

(c) Hoeveel grade per minuut word gedek wanneer die bal teen 8.00 rev/s draai?



$$r_1 = 0.600 \text{ m} \rightarrow 8.00 \text{ rev/s}$$

$$r_2 = 0.900 \text{ m} \rightarrow 6.00 \text{ rev/s}$$

We need to determine the speeds from both these rates of rotation.

$$v = \frac{2\pi r}{T} \text{ where } T = \frac{1}{f} \text{ } \left(\begin{array}{l} \text{period and} \\ \text{frequency or rate of rot.} \end{array} \right)$$

(a)

$$\therefore v_1 = 2\pi r_1 f_1 \text{ with } f_1 = 1/T_1$$

$$= 2\pi(0.600 \text{ m})(8.00 \text{ rev/s})$$

$$= 2(3.14)(0.600 \text{ m})(8.00 \text{ rev/s}) = 30.2 \text{ m/s}$$

$$v_2 = 2(3.14)(0.900 \text{ m})(6.00 \text{ rev/s}) = 33.9 \text{ m/s}$$

(b) $a_{c1} = \frac{v_1^2}{r_1} = \frac{(30.2 \text{ m/s})^2}{0.600 \text{ m}} = 1.52 \times 10^3 \text{ m/s}^2$ - 1/2 for units or sig figs and figures

and $a_{c2} = \frac{v_2^2}{r_2} = \frac{(33.9 \text{ m/s})^2}{0.900 \text{ m}} = 1.28 \times 10^3 \text{ m/s}^2$

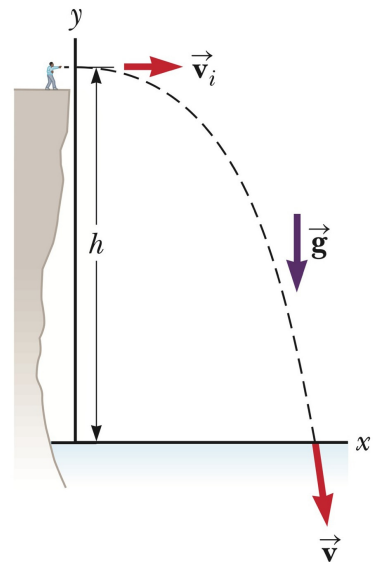
(c) $8.00 \text{ rev/s} = 8.00 \text{ rev} \left(\frac{360^\circ}{1 \text{ rev}} \right)$ (3)

$$\frac{1 \text{ sec} \left(\frac{1 \text{ min}}{60 \text{ sec}} \right)}{1 \text{ sec}} = (1.73 \times 10^5)^\circ \text{ per minute}$$

Question 5 / Vraag 5

A student stands at the edge of a cliff and throws a stone horizontally over the cliff with a speed of 18.0 m/s. The cliff is $h = 50.0$ m above a body of water, as shown in the accompanying figure. [20]

- What are the co-ordinates of the initial position of the stone? (2)
- What are the components of the initial velocity of the stone? (2)
- What is the appropriate analysis model for the vertical motion of the stone? (2)
- What is the appropriate analysis model for the horizontal motion of the stone? (2)
- Write symbolic equations for the x and y components of the velocity of the stone as a function of time. (2)
- Write symbolic equations for the x and y components of the position of the stone as a function of time. (2)
- How long after being released does the stone strike the water at the bottom of the cliff? (3)
- With what speed and angle of impact (relative to the horizontal) does the stone hit the water? (5)



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'n Student staan aan die rand van 'n krans en gooi 'n klip horisontaal oor die krans teen 'n spoed van 18.0 m/s. Die krans is $h = 50.0$ m bokant 'n massa water, soos in die meegaande figuur aangetoon.

- Wat is die koördinate van die beginposisie van die klip?
- Wat is die komponente van die beginsnelheid van die klip?
- Wat is die toepaslike ontledingsmodel vir die vertikale beweging van die klip?
- Wat is die toepaslike ontledingsmodel vir die horisontale beweging van die klip?
- Skryf simboliese vergelykings vir die x- en y-komponente van die snelheid van die klip as 'n funksie van tyd.
- Skryf simboliese vergelykings vir die x- en y-komponente van die posisie van die klip as 'n funksie van tyd.

(g) Met watter spoed en invalshoek slaan die klipgrond?

(g) $V_i = 18.0 \text{ m/s}$, $h = 50.0 \text{ m}$ – given parameters.

(a) Co-ordinates of the initial position of the stone:

$$x_i = 0.00 \text{ m}, y_i = 0.00 \text{ m}$$

(2)

(b) Components of initial velocity:

$$v_{xi} = 18.0 \text{ m/s}, v_{yi} = 0.0 \text{ m/s}$$

(2)

(c) Free fall motion, with constant downwards acceleration, $g = 9.80 \text{ m/s}^2$

(2)

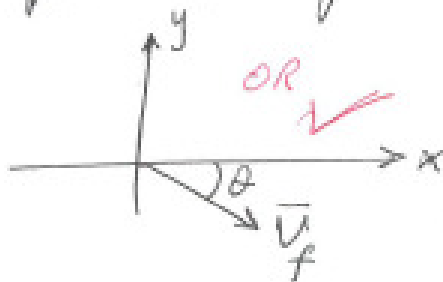
Question 5 continued

$$\begin{aligned}\therefore V_f &= \sqrt{(V_{xf})^2 + (V_{yf})^2} \\ &= \sqrt{(18.0 \text{ m/s})^2 + (-31.3 \text{ m/s})^2} \\ &= 36.1 \text{ m/s} \checkmark\end{aligned}$$

$$\begin{aligned}\text{and } \theta_f &= \tan^{-1}\left(\frac{V_{yf}}{V_{xf}}\right) \\ &= \tan^{-1}\left(\frac{-31.3 \text{ m/s}}{18.0 \text{ m/s}}\right)\end{aligned}$$

(5)

$= -60.1^\circ$, which means that the velocity points in the 4th quadrant since its y-component is negative.

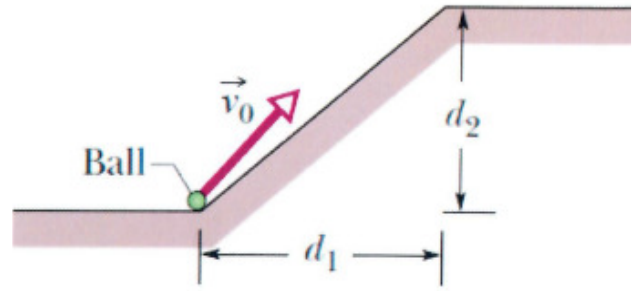


Question 6 / Vraag 6

In the accompanying figure, a ball is launched with a velocity of 10.0 m/s, at an angle of 50.0° to the horizontal. The launch point is at the base of a ramp of horizontal length $d_1=6.00$ m and height $d_2 = 3.60$ m. A plateau is located at the top of the ramp [12]

(a) Assume that the ball lands on the ramp and calculate the x coordinate of the point at which the ball hits the ramp. **HINT:** Determine the gradient of the ramp and then write the height of the ramp, y_r , as a function of x . (10)

(b) When the ball lands, what is the magnitude of its displacements from the launch point? (2)

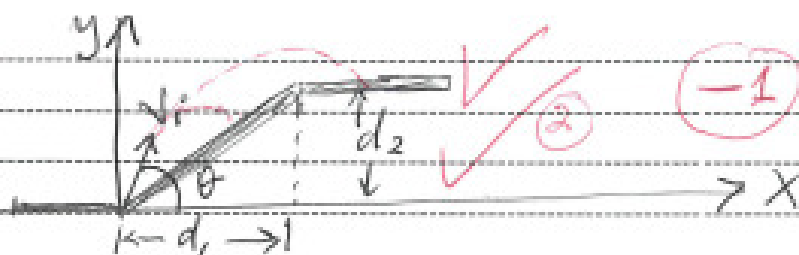


In die meegaande figuur word 'n bal lanseer met 'n snelheid van 10.0 m/s , teen 'n hoek van 50.0° tot die horisontaal. Die beginpunt is aan die voet van 'n oprit met horisontale lengte $d_1 = 6.00 \text{ m}$ en hoogte $d_2 = 3.60 \text{ m}$. Aan die bokant van die oprit is 'n Plato geleë.

(a) Neem aan dat die bal op die oprit land en bereken die x -koordinaat van die punt waar die bal die oprit tref. **WENK:** Bepaal die gradient van die oprit en skryf dan die hoogte van die oprit, y_r , as 'n funksie van x .

(b) Wanneer die bal land, wat is die grootte van die verplasing vanaf die beginpunt?

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.



① No axes, labels

The origin is placed at the launch point.

$v_i = 10.0 \text{ m/s}$ and $\theta = 50.0^\circ$ — some given param

In the horizontal direction:

In order for the ball to land on the plateau,

$$x_f > 6.00 \text{ m}$$

$$\Rightarrow x_f = v_{xi} t + \frac{1}{2} a_x t^2 \quad \text{with } x_i = 0 \text{ and } a_x = 0 \quad \text{(at launch point)}$$

$$\therefore t = \frac{x_f}{v_{xi}} = \frac{x_f}{v_i \cos \theta} \quad \text{--- (1)}$$

In the vertical direction:

$$y_f - y_i = v_{yi} t + \frac{1}{2} a_y t^2 \quad \text{--- (2)}$$

$$y_f = (v_i \sin \theta) t - \frac{1}{2} a_y t^2 \quad \text{since } y_i = 0 \text{ at launch point}$$

$$= (v_i \sin \theta) t - \frac{1}{2} (9.80 \text{ m/s}^2) t^2$$

Substituting t from eqn (1) into eqn (2):

$$y_f = v_i \sin \theta \left(\frac{x_f}{v_i \cos \theta} \right) - \frac{1}{2} (9.80 \text{ m/s}^2) \left(\frac{x_f}{v_i \cos \theta} \right)^2 \quad \text{--- (3)}$$

A relationship between x and y can be established from the gradient of the ramp:

$\therefore y = mx + c$ with $c = 0$ (at the launch point)

$$m = \frac{\Delta y}{\Delta x} = \frac{d_2}{d_1} = \frac{3.60 \text{ m}}{6.00 \text{ m}} = 0.60$$

$$\therefore y = 0.60x \quad \text{is the required relation} \quad \text{--- (4)}$$

Substituting eqn (4) into (3):

with $y_f = y$ and $x_f = x$

$$\therefore y_f = V_i \sin \theta \left(\frac{x_f}{V_i \cos \theta} \right) - \frac{1}{2} (9.80 \text{ m/s}^2) \left(\frac{x_f}{V_i \cos \theta} \right)^2$$

$$0.60 x_f = (\tan \theta) x_f - \frac{1}{2} (9.80 \text{ m/s}^2) \left(\frac{x_f}{V_i \cos \theta} \right)^2$$

$$0.60 x_f = x_f \tan \theta - 4.9 \frac{x_f^2}{V_i^2 \cos^2 \theta}$$

$$0.60 = \tan \theta - 4.9 \frac{x_f}{V_i^2 \cos^2 \theta}$$

$$\therefore x_f = (0.60 - \tan(50.0^\circ)) \frac{V_i^2}{4.9 \cos^2 50^\circ}$$
$$= 4.99 \text{ m}$$

(b) The x co-ordinate for where the ball lands is already determined.

The y co-ordinate is easily determined from the slope:

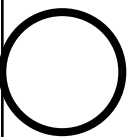
$$y = mx = 0.60 (4.99 \text{ m})$$
$$= 2.994 \text{ m}$$

The ball just misses the platform.

$$r = (x_f^2 + y_f^2)^{1/2} = [(4.99 \text{ m})^2 + (2.994 \text{ m})^2]^{1/2} = 5.82 \text{ m}$$

$$\theta = \tan^{-1} \left(\frac{2.994}{4.99} \right) = 30.1^\circ \checkmark$$

Handwriting practice lines consisting of multiple horizontal dashed lines for tracing and writing practice.

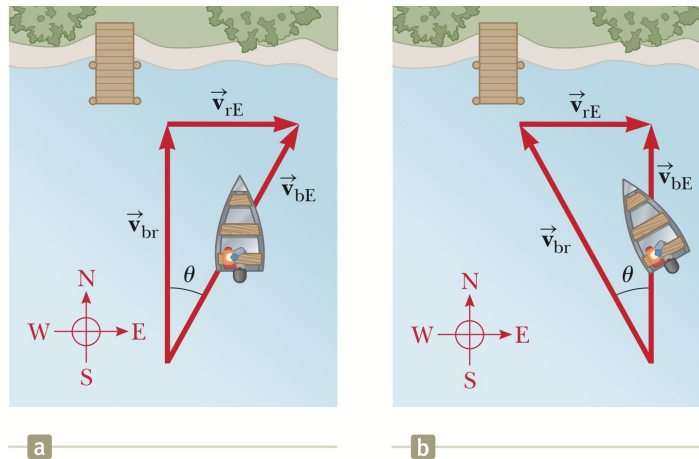


Question 7 / Vraag 7

A boat crossing a wide river moves with a speed of 10.0 km/h relative to the water. The water in the river has a uniform speed of 5.00 km/h due east relative to the earth. [6]

(A) If the boat heads due north, determine the velocity of the boat relative to an observer standing on either bank. (3)

(B) If the boat travels with the same speed of 10.0 km/h relative to the river and is to travel due north (as shown in part (b) of the accompanying figure), what should its heading be (ie. where should the skipper point it in order to travel due north)? (3)



'n Boot wat 'n wye rivier oorsteek, beweeg met 'n spoed van 10 km/h relatief tot die water. Die water in die rivier het 'n uniforme spoed van 5,00 km/h oos relatief tot die aarde.

(A) As die boot noordwaarts stuur, bepaal die snelheid van die boot relatief tot 'n waarnemer wat op enige oewer staan.

(B) As die boot met dieselfde spoed van 10,0 km/h relatief tot die water vaar, maar reg noord moet beweeg, (soos aangedui in deel (b) van die meegaande figuur), in watter rigting moet die skipper die boot stuur?

Handwritten solution for Question 7:

(A) $\vec{v}_{be} = \vec{v}_{br} + \vec{v}_{re}$ ✓
 Find speed v_{be} :

$$v_{be} = \sqrt{(v_{br})^2 + (v_{re})^2} = \sqrt{(10.0 \text{ km/h})^2 + (5.00 \text{ km/h})^2}$$

$$= 11.2 \text{ km/h} \quad \checkmark$$

$\theta = \tan^{-1}\left(\frac{v_{re}}{v_{br}}\right) = \tan^{-1}\left(\frac{5.00 \text{ km/h}}{10.0 \text{ km/h}}\right)$ (3)
 $= 26.6^\circ \quad \checkmark$

(B) $\vec{v}_{br} = \vec{v}_{re} + \vec{v}_{be}$ ✓

$$v_{be} = \sqrt{(v_{br})^2 - (v_{re})^2} = \sqrt{(10.0 \text{ km/h})^2 - (5.00 \text{ km/h})^2}$$

$$= 8.66 \text{ km/h} \quad \checkmark$$
 (3)

END / EINDE
 Good Luck! / Veels geluk!

$\theta = \tan^{-1}\left(\frac{v_{re}}{v_{be}}\right)$
 $= \tan^{-1}\left(\frac{5.00}{8.66}\right)$
 $= 30.0^\circ \quad \checkmark$

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